



NATIONAL UNIVERSITY OF MODERN LANGUAGES

SOFTWARE QUALITY ENGINEERING LAB REPORT – 5

NAME: Abdul Rafay

ROLL NO: FL23791

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COURSE: SOFTWARE QUALITY ENGINEERING

SUBMITTED TO: Mam Rubab

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Software Metrics and GQM Model Report

Task 1 — Software Metrics Classification

Selected Project: Inventory Management System

An Inventory Management System is used by businesses to manage stock, suppliers, sales, purchases, and warehouse operations.

A. Product Metrics

Metric Name	Hypothetical Value	Unit of Measurement
Lines of Code (LOC)	45,000	Lines
Number of Modules	18	Modules
Defect Density	2.5	Defects/KLOC
System Response Time	2.1	Seconds
Availability	99.5	Percentage
User Satisfaction Rate	88	Percentage
Database Size	120	GB
Number of Functions	75	Functionalities
Reliability Rate	98.7	Percentage
Transaction Processing Speed	350	Transactions/Minute

B. Process Metrics

Metric Name	Hypothetical Value	Unit of Measurement
Development Time	8	Months
Average Bug Fix Time	4	Hours
Testing Coverage	85	Percentage
Number of Developers	10	Team Members
Defects Found During Testing	140	Defects

Requirements Change Requests	22	Requests
Code Review Completion Rate	92	Percentage
Sprint Completion Rate	90	Percentage
Build Success Rate	95	Percentage
Customer Support Resolution Time	6	Hours

Product metrics focus on the quality and performance of the final software product, while process metrics focus on the efficiency of the software development process.

Task 2 — GQM Model for SmartBank Digital Banking Application

SmartBank is a digital banking application that supports account balance checking, fund transfers, bill payments, loan applications, mobile top-ups, and transaction history review. The system supports both web and mobile platforms and serves more than 50,000 active users.

Goal 1: Improve Reliability

The goal is to reduce transaction failures and system downtime during banking operations.

Question	Metric	Unit
How often does the system fail during transactions?	Transaction Failure Rate	Percentage
How much downtime occurs monthly?	System Downtime	Hours/Month
How stable is the application during peak hours?	Server Crash Count	Number of Crashes
How quickly are failures resolved?	Mean Time to Repair (MTTR)	Minutes
How many critical defects are reported?	Critical Bug Count	Number of Bugs
How reliable are transaction operations?	Successful Transaction Rate	Percentage
How fast does the system respond?	Average Response Time	Seconds

Goal 2: Improve Usability

The goal is to make the application easier for users to understand and use efficiently.

Question	Metric	Unit
Do users understand navigation easily?	User Navigation Error Rate	Percentage
How many complaints are received?	Customer Complaint Count	Complaints/Month
How easy is it to complete transactions?	Task Completion Rate	Percentage
How much time is needed to complete a task?	Average Task Completion Time	Minutes
Are users satisfied with the interface?	User Satisfaction Score	Rating out of 5
How many help requests are generated?	Help Request Frequency	Requests/Week
How many users leave before completion?	Transaction Abandonment Rate	Percentage

Benefits of GQM Model

- Measure software quality using real data
- Reduce transaction failures and downtime
- Improve customer satisfaction
- Support better decision making
- Improve software testing and maintenance
- Enhance system reliability and usability

Conclusion

The GQM model helps SmartBank define measurable software quality goals and evaluate system performance using proper metrics. This approach supports reliability, usability, and continuous improvement of the banking application.